



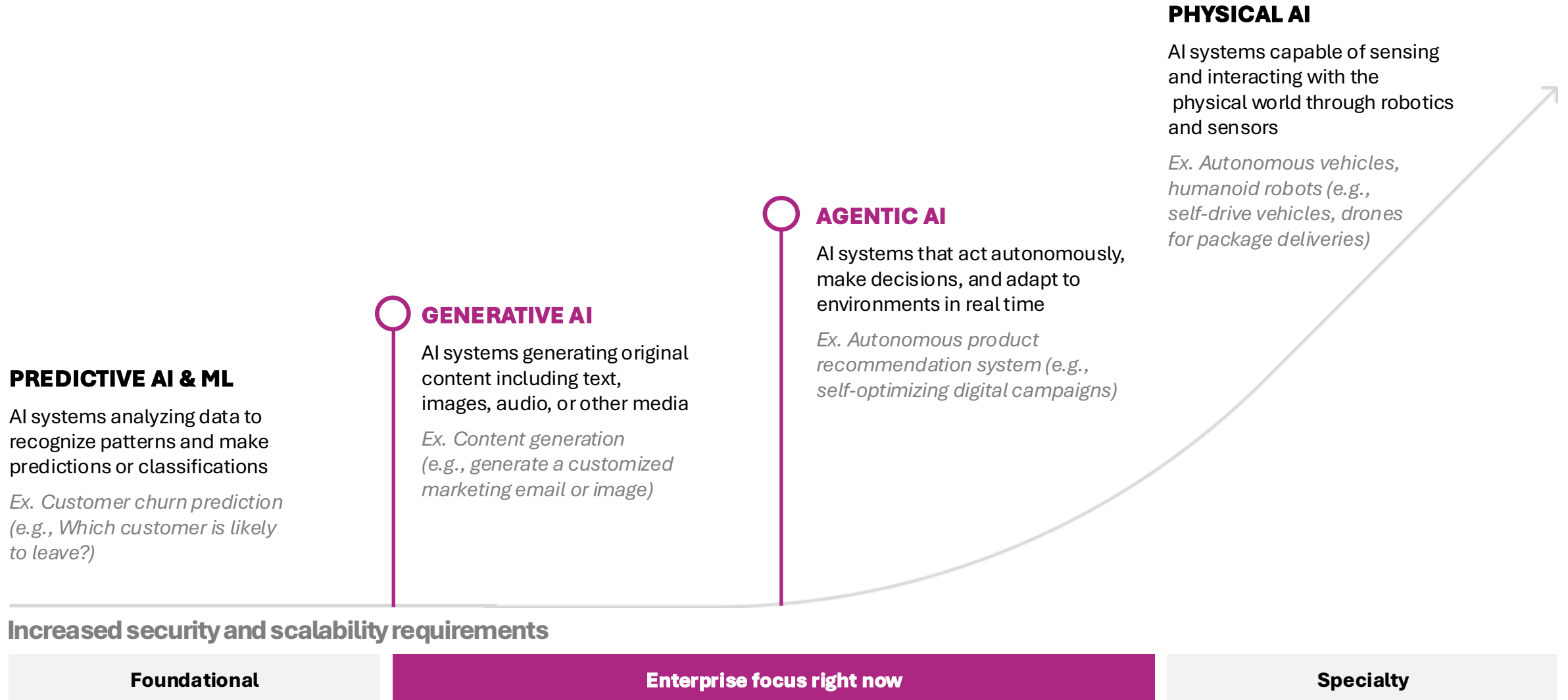
F5 Inference Optimizer: Maximizing Token Efficiency

Presenter Name (pronouns)

Title

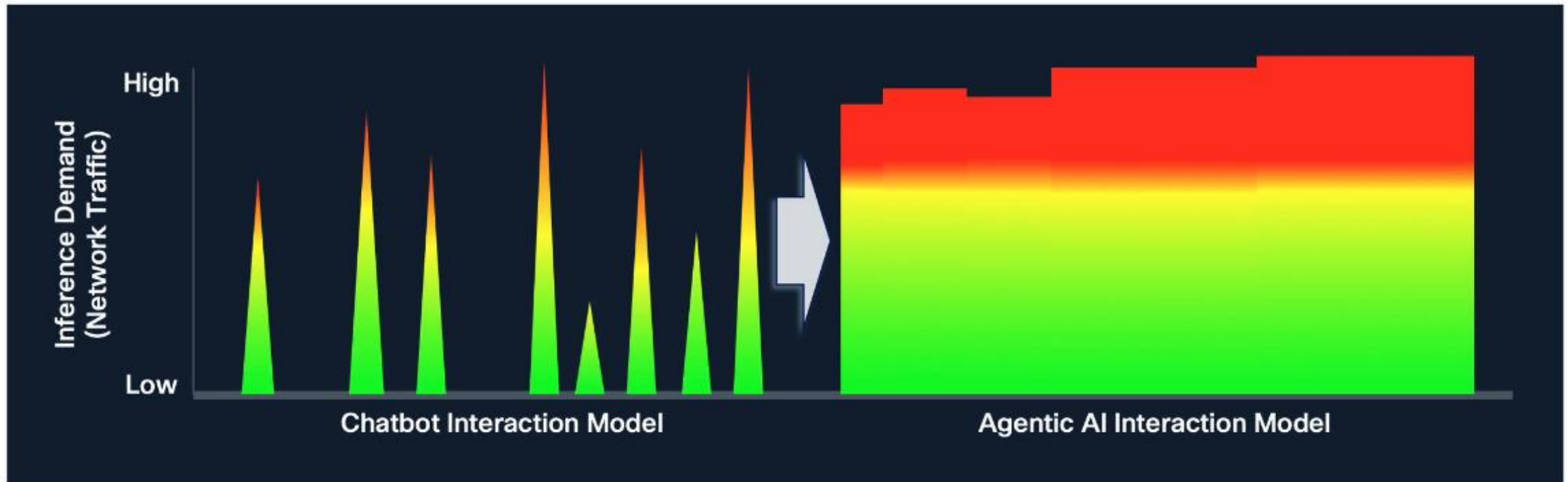
Date

AI is advancing, reshaping critical elements of our lives



A new level of AI demand from the infrastructure that requires different solution

Ingress, egress, east/west, security, etc.



This graphic is for illustrative purposes only. It is not based on customer network data but depicts how Cisco believes network traffic levels will increase as inference demand further increases and proliferates within enterprise and end-user networking environments.

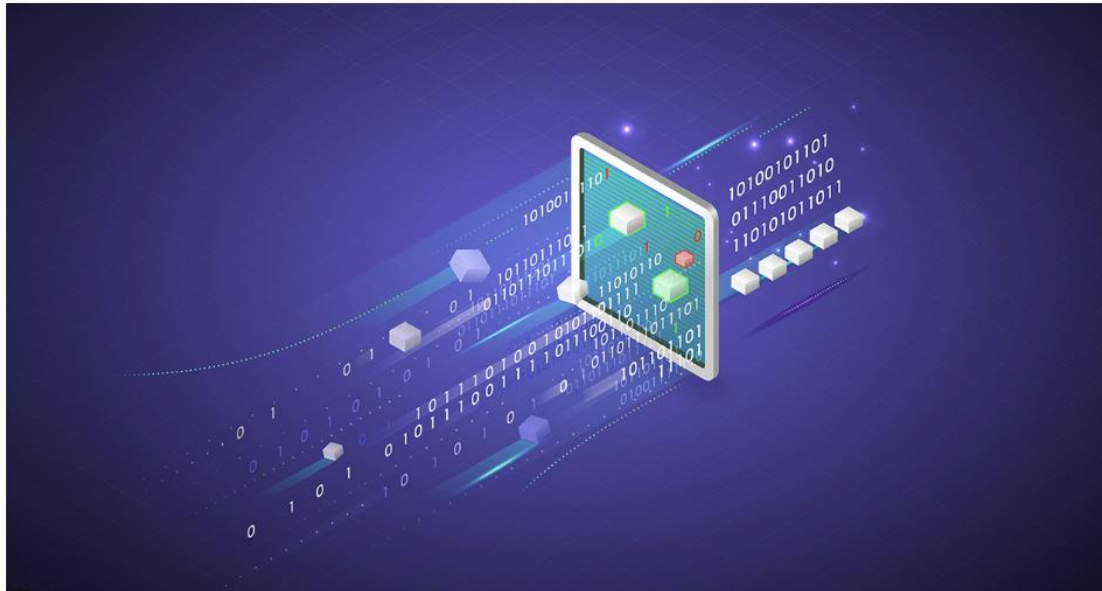
All need very high performance

Token Economy - Tokenomic

Explaining Tokens — the Language and Currency of AI

Tokens are units of data processed by AI models during training and inference, enabling prediction, generation and reasoning.

March 17, 2025 by [Dave Salvator](#)



“

This is an industrial revolution, because for the first time, we’re producing something entirely new at extremely high volume — floating point numbers, or **tokens**. These tokens have value because they represent **artificial intelligence**.

You can take these floating point numbers and reformulate them into language, proteins, chemicals, images, robotic controls — we’ve discovered how to produce tokens of almost any kind at extraordinary scale. The world is going to produce an **enormous amount** of these AI tokens.

These tokens will be produced in new types of data centers we call **AI factories**. Electrons go in, and floating point numbers come out. Just like how electricity was sold per kilowatt-hour in the last industrial revolution, now we’ll have millions of AI tokens per dollar.”

Key focus areas for companies investing and deploying AI infrastructure



Maximize Investment

Ensure that valuable GPU and CPU resources are fully utilized

Increase the number of supported users and customers

Security

Secure data, apps, and APIs
Isolate user and customers on shared infrastructure

Agility and Scale

Move from POCs to production with full stack integration with observability

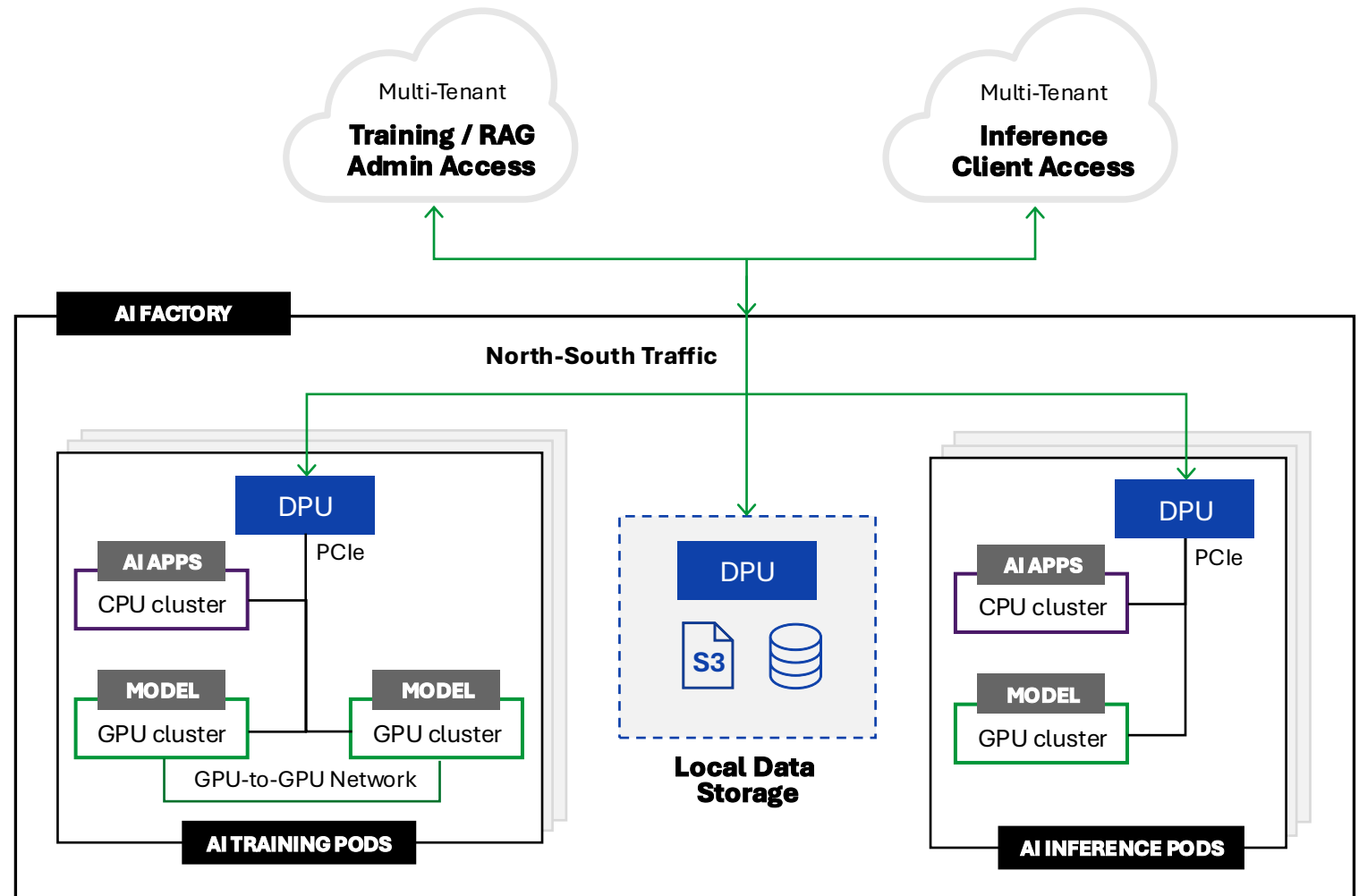
Agility to keep pace with quickly evolving AI use case

AI Factories: Next generation data centers

AI infrastructure differs from traditional data center architectures, **driven by unique AI workload** requirements

DPUs are becoming a key component in accelerated networking for AI infrastructure

Kubernetes is the de facto standard for AI workload deployments, confronting scaling and complexity challenges that **F5 solves**



Standard HTTP vs AI Inferencing

Standard HTTP Workloads

(Round-robin works well)

- **Request duration:** predictable latency
- **Payload size:** consistent
- **Compute cost per request:** equal
- **Queueing model:** stateless, even queue (FIFO)
- ✓ **Round-robin result:** even server utilization

Near- perfect balance achieved

AI inference workloads

(Round-robin fail)

- **Request duration:** huge latency variance
- **Payload size:** huge size variance
- **GPU compute cost:** depends on prompt and context
- **Queueing model:** stateful KV-cache queue
- ✓ **Round-robin result:** GPU queue imbalance (~30% GPU waste, P99 latency spikes)

High cost and latency

The GPU-to-output conversion problem

- **GPU compute accounts for ~40–60% of AI infrastructure budgets**

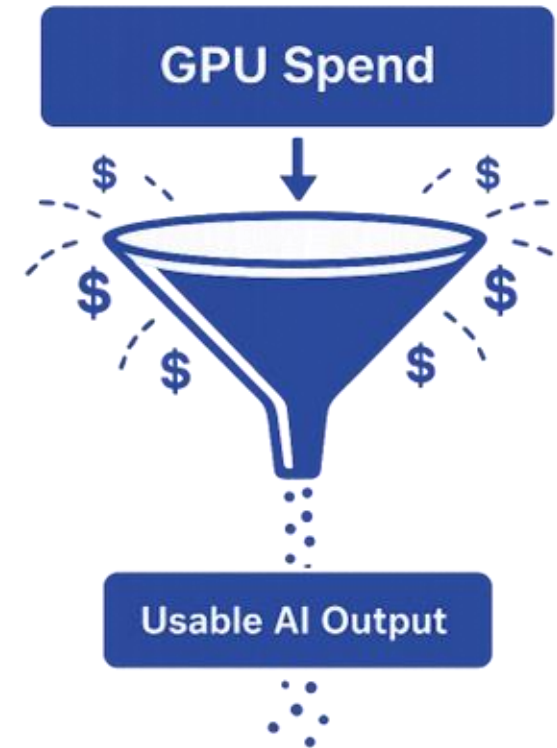
Source: *arXiv.org – AI infrastructure cost analysis*

- **A single underutilized GPU can waste ~\$70,000 per year**

Source: *Kube.Today / vCluster analysis*

- **GPU cost optimization is the #1 challenge in AI infrastructure**

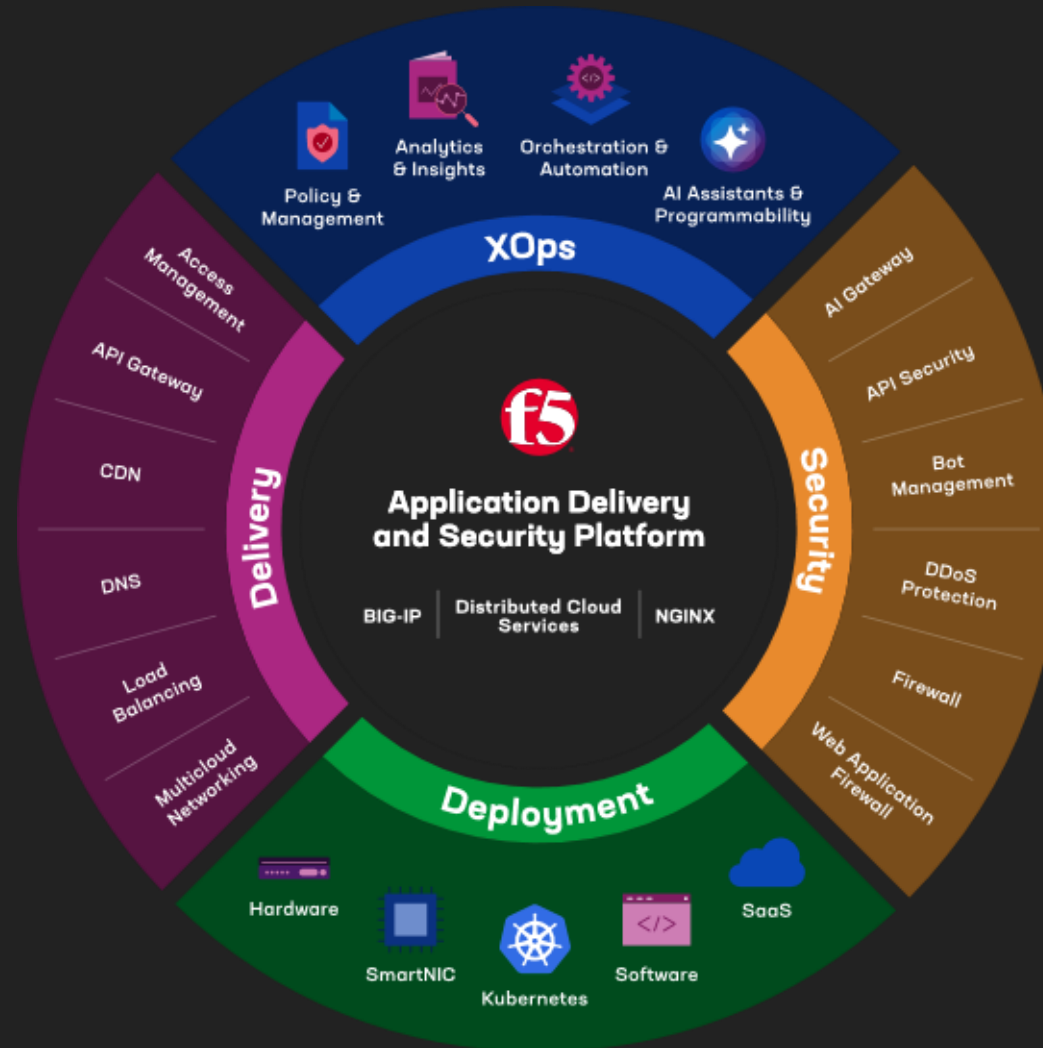
Source: *Kube.Today survey*



Sources: Kube.Today (2025 AI Infrastructure Survey); vCluster analysis; arXiv AI infra cost studies; Reuters & Bloomberg on NVIDIA AI GPU pricing; SemiAnalysis on inference economics.

F5 Application Delivery and Security Platform (F5 ADSP)

Consolidates all app & API delivery and security for today's hybrid, multicloud and AI driven era



F5 enterprise AI delivery and security solutions offers a unified, scalable solution to protect and optimize AI workloads



F5 leverages decades of application delivery expertise with enterprise-grade AI security to optimize performance, scalability, and protection for AI workloads.



AI data delivery

Ensure fast, secure, and reliable data pipelines to fuel AI model training at scale.



AI factory load balancing

Minimize costs and maximize GPU utilization while streamlining AI workloads for efficient infrastructure.



AI runtime security

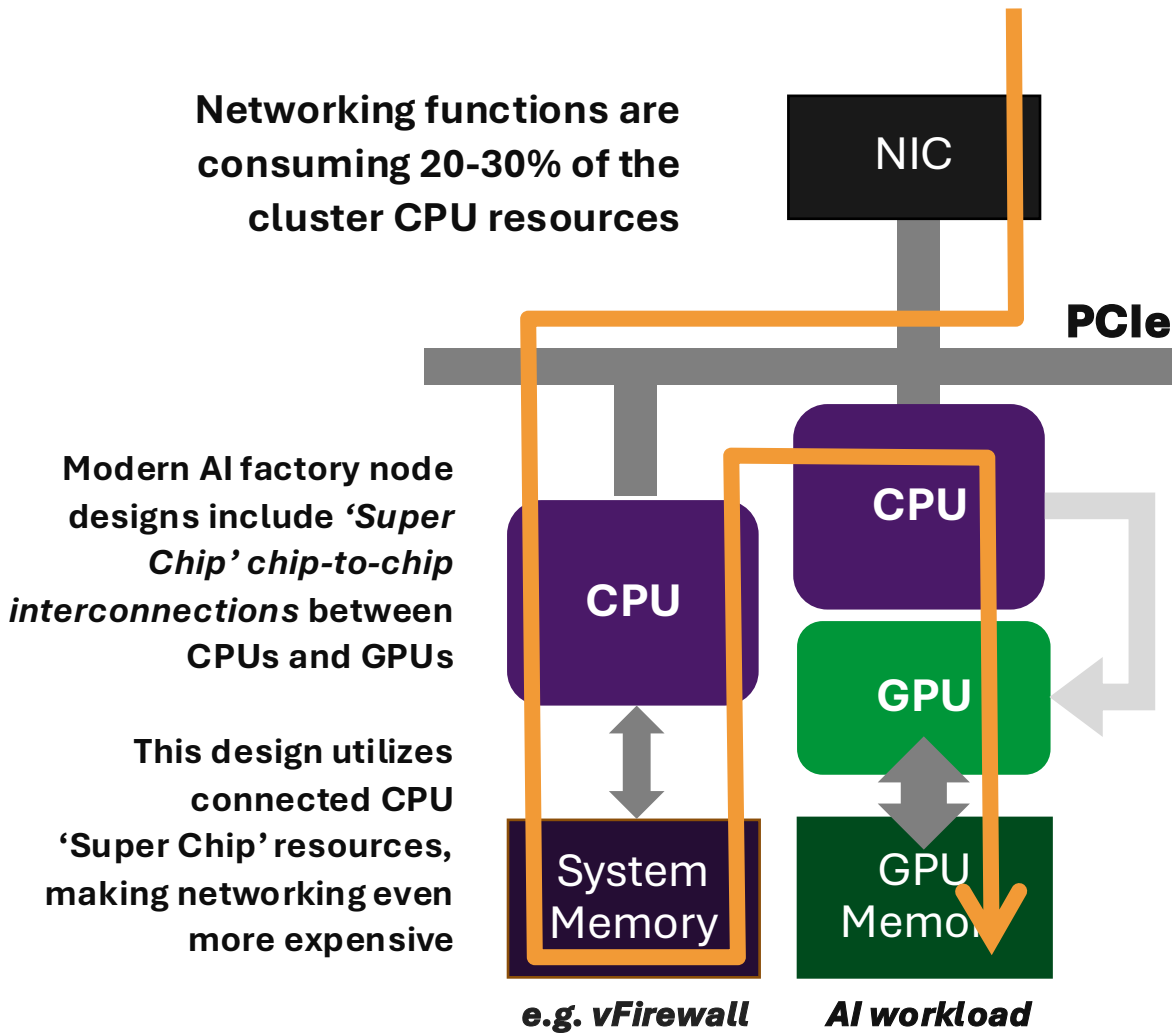
Safeguard AI applications, APIs, and models against emerging threats like data leakage, API exploitation, shadow AI, and malicious injections.



AI multicloud networking

Enable seamless and secure data and model connectivity across distributed AI environments.

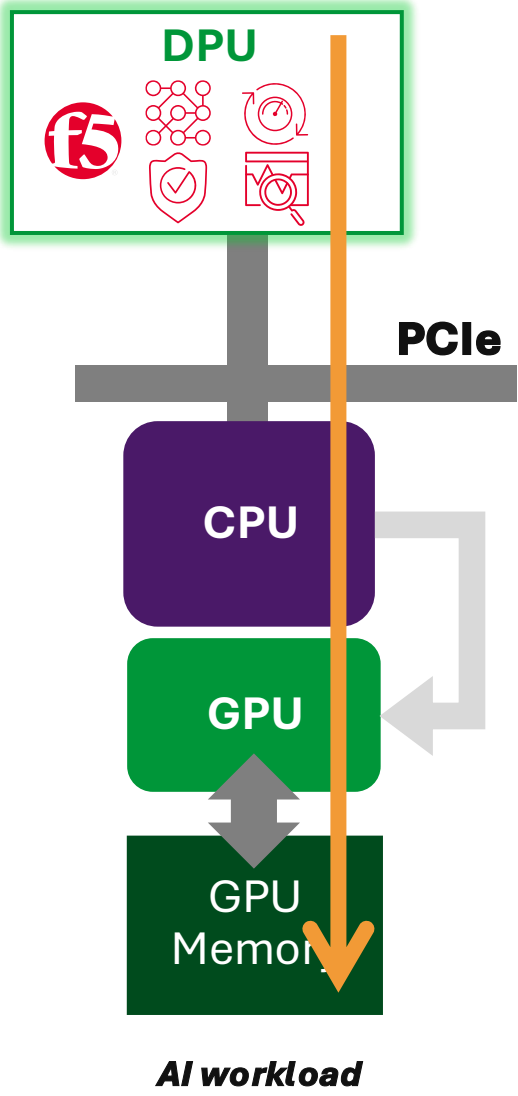
BIG-IP Next services bring innovation offered inline for maximum efficiency



AI factories are increasingly requiring DPU deployments for the use of hardware ‘accelerators’ for inter-node (East-West) traffic

Required DPU usage represent a market for BIG-IP software

BIG-IP Next will take advantage of these accelerators to keep up with increasing networking capacities



Leveraging F5 BIG-IP Next for Kubernetes to Solve AI Factory Networking Challenges

BIG-IP Next for Kubernetes drives higher GPU ROI, stronger tenant security, predictable token spend, and easy full-DPU adoption

Programmable AI Traffic Optimization

Value: Higher GPU ROI, faster responses, better SLA adherence

- Intelligent AI LB
- LLM Routing
- Semantic Model Caching
- Programmability via iRules

Multi-tenancy & Zero-trust Security

Value: Tenants are isolated, attackers stopped at the boundary

- Network-based Multi-tenancy
- Per-tenant ingress/egress ACL
- DPU-offloaded Security
- Secure MCP
- Tenant-Aware Service Segmentation

Token Analytics & Cost Governance

Value: Real-time token usage and model performance visibility per tenant

- Token Usage Reporting
- Tenant-aware Metrics
- Real-time Model Performance Insights
- Tenant RBAC
- Multi-cluster visibility

Simplified Deployment & Lifecycle Operations

Value: Full DPU benefits without operational complexity

- NVIDIA DPF Integration
- Future Integrations:
 - Rafay
 - Nvidia run:ai
- F5 Lifecycle Operator (FLO)
- K8s Gateway API

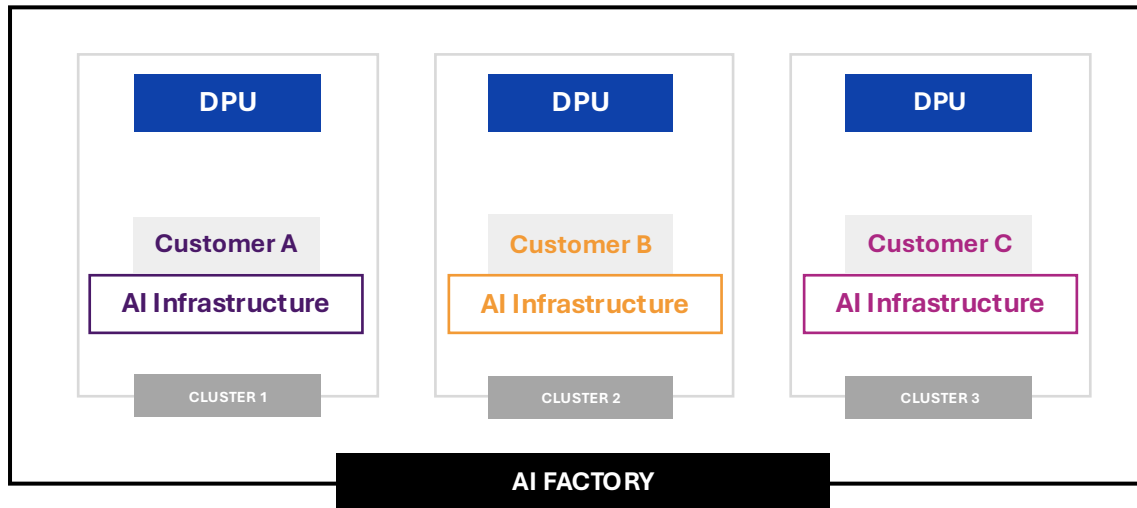
Use Cases and Capabilities

**Unlock the Full Monetization
Potential of AI Infrastructure**

Multi-tenancy and security are critical requirements for AI-aaS and GPU-aaS providers to maximize ROI

Without

BIG-IP Next for Kubernetes

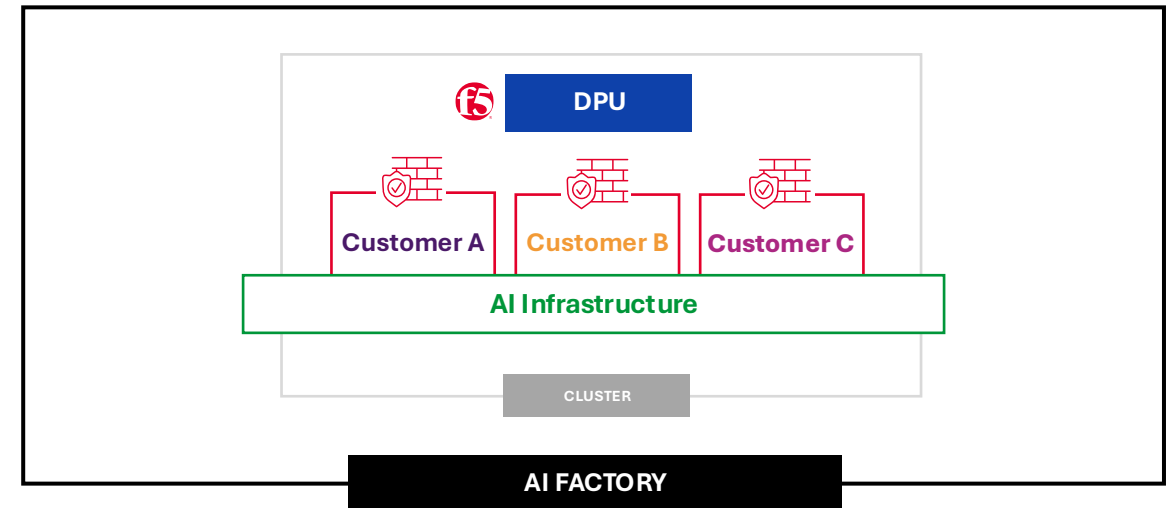


Underutilization

To achieve multi-tenancy AI-aaS and GPU-aaS providers dedicate clusters per customer

With

BIG-IP Next for Kubernetes



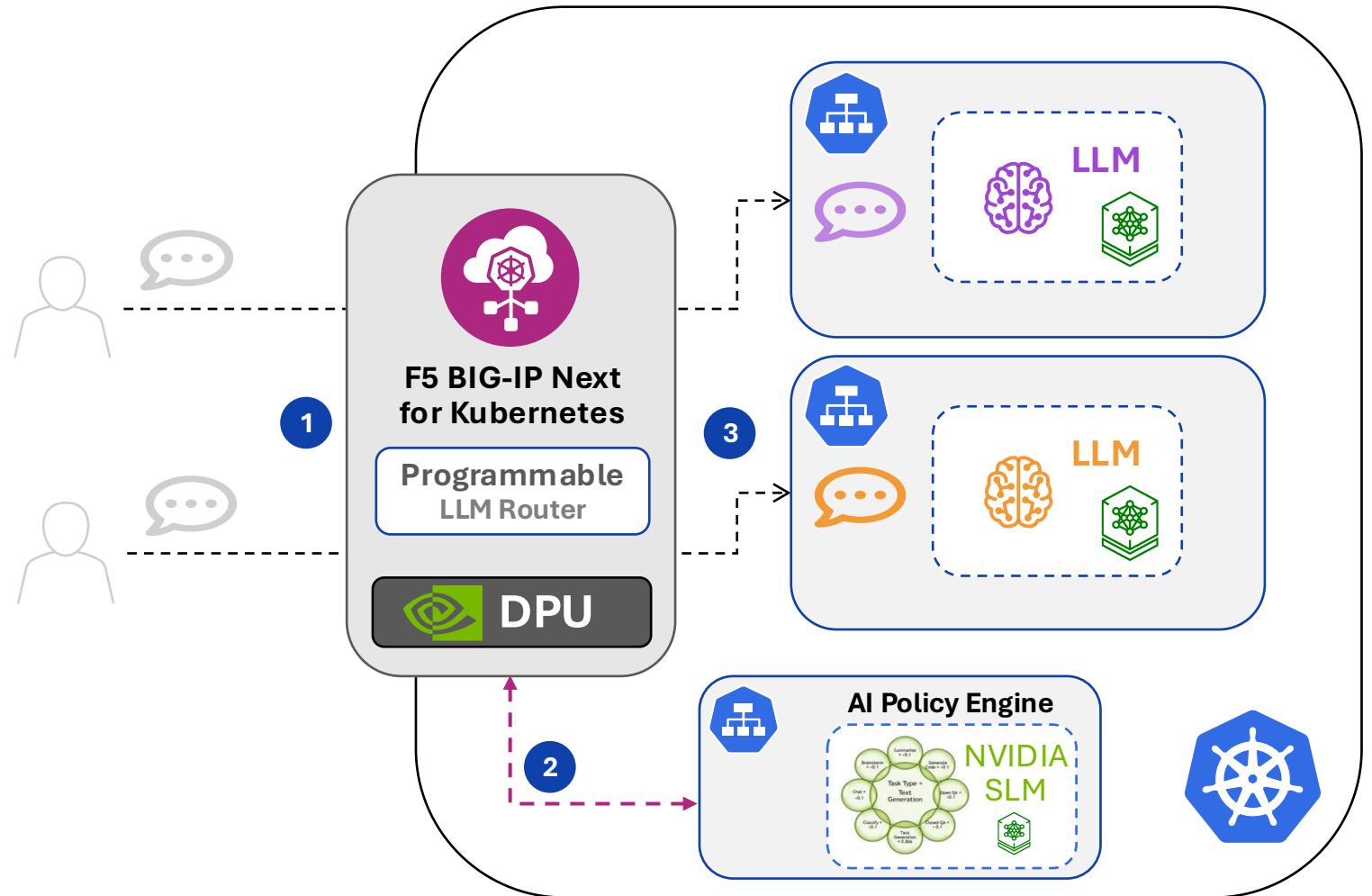
Resource Maximization

BIG-IP Next for Kubernetes allows customers' tenants to be isolated for security on shared AI infrastructure

LLM Routing – Routing AI prompts to the right place for the right outcome

LLM Router = BIG-IP for K8s Programmable Data Plane combined with NVIDIA NIM

- 1 Users send prompts which are intercepted based on BIG-IP for K8s programmable business logic
- 2 BIG-IP for K8s routes prompt requests to NVIDIA NIM for classification
- 3 Based on classification from NVIDIA NIM BIG-IP for K8s routes the prompt to the best LLM



Models Are Made Different

Models are different in terms of strength. Matching prompts to specialized models ensure better accuracy, relevancy and response time

Smaller Model



- Simple prompt
- Basic Q&A
- Cheaper to run
- Faster response time

e.g. What's the capital city of Australia?

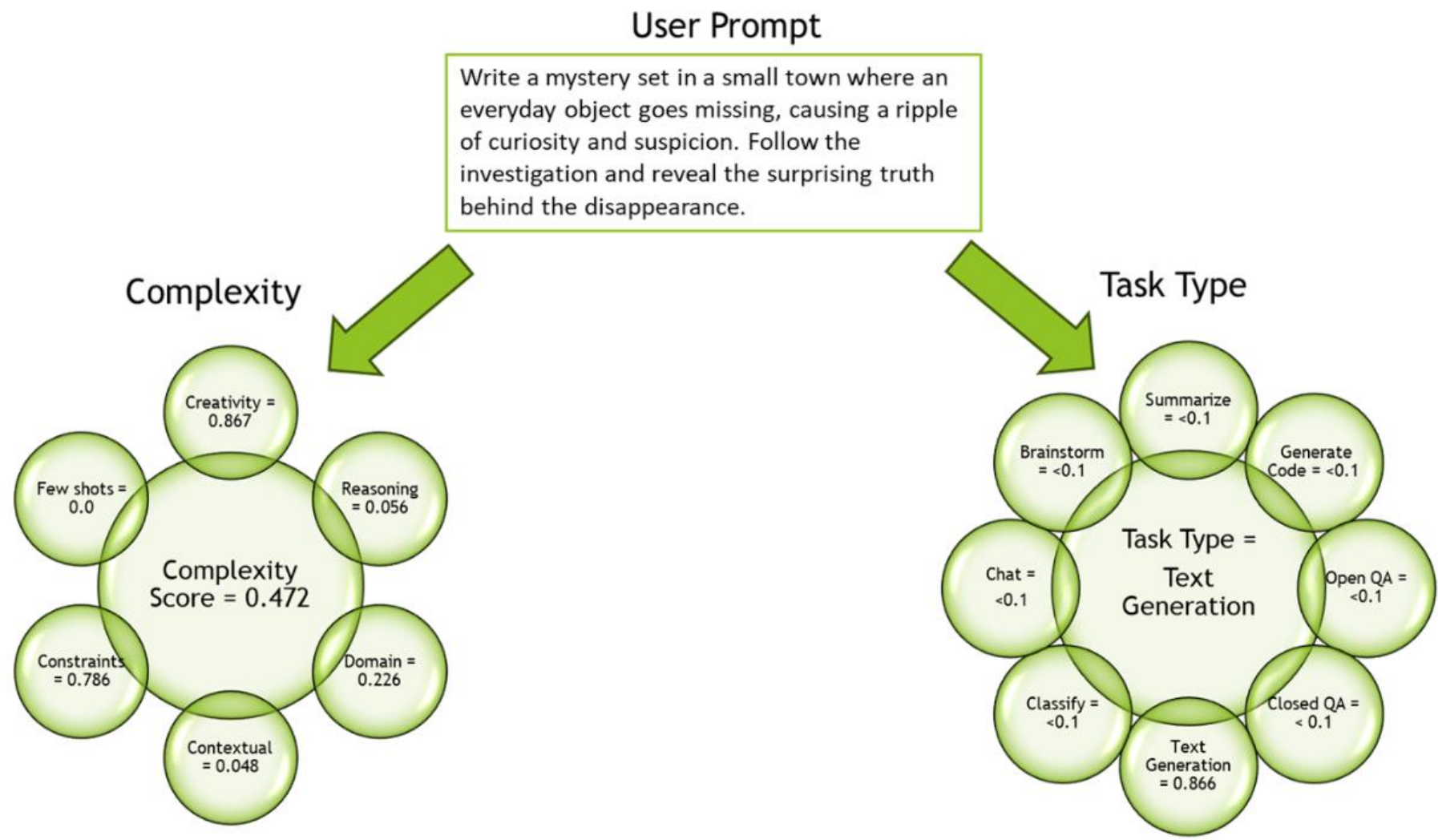
Larger Model



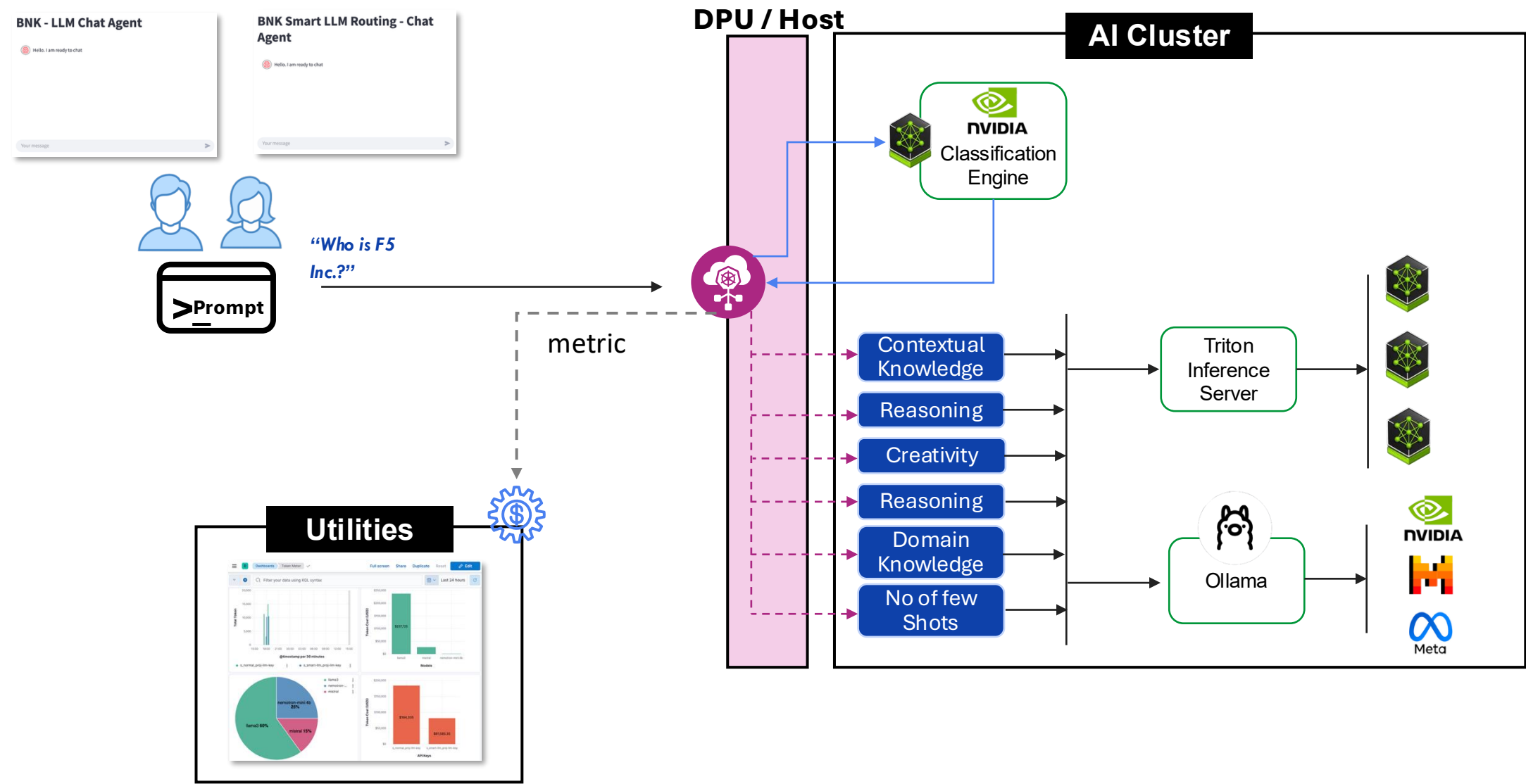
- Complex prompt
- Multi-step reasoning
- Domain knowledge
- Computational expensive
- Consume more power per token generation

e.g. Help me to plan a trip from Australia to USA and give me the best air fair.

NVIDIA Prompt Task and Complexity Classifier



How Smart LLM Routing works?



LLM Load Balancing – significantly lower time to first token, higher tokens generation, and ultimately greater prompt throughput

Optimized Inferencing with greater GPU utilization:

Intelligent and programable LLM Load Balancing enabled by BIG-IP for K8s + NVIDIA

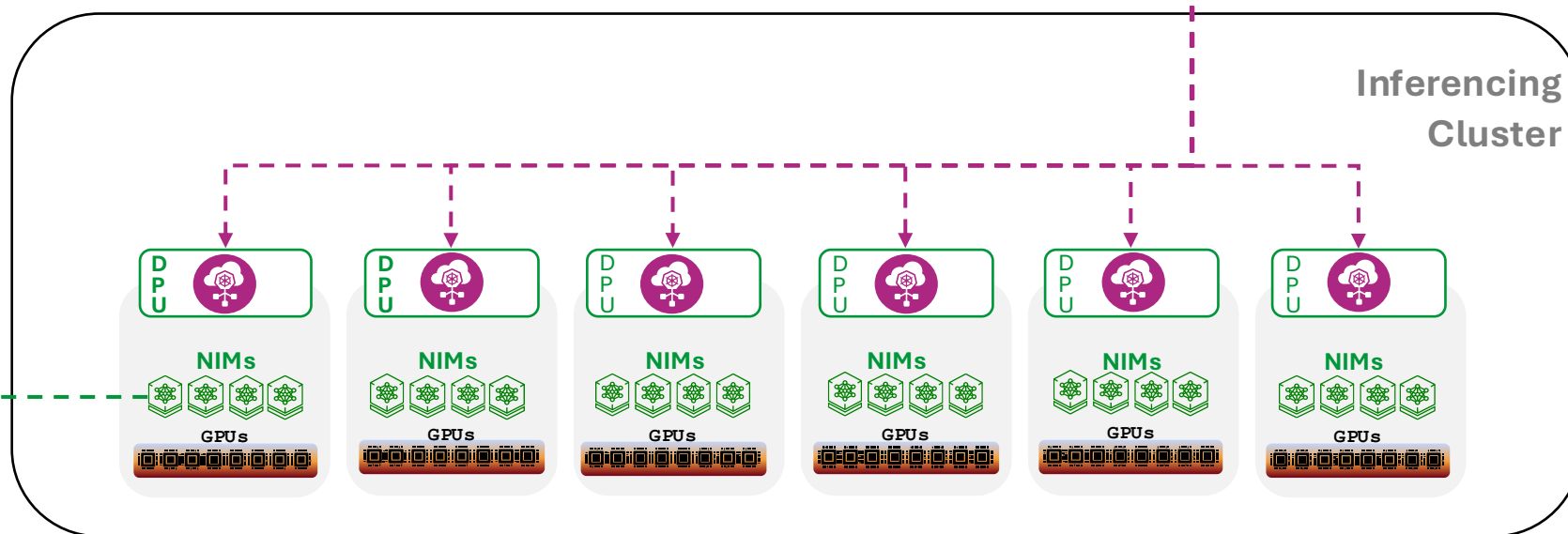


- 1 Telemetry data from Inference Cluster is used to update BIG-IP for K8s load balancing rules
- 2 New load balancing rules are applied to optimize inference utilization

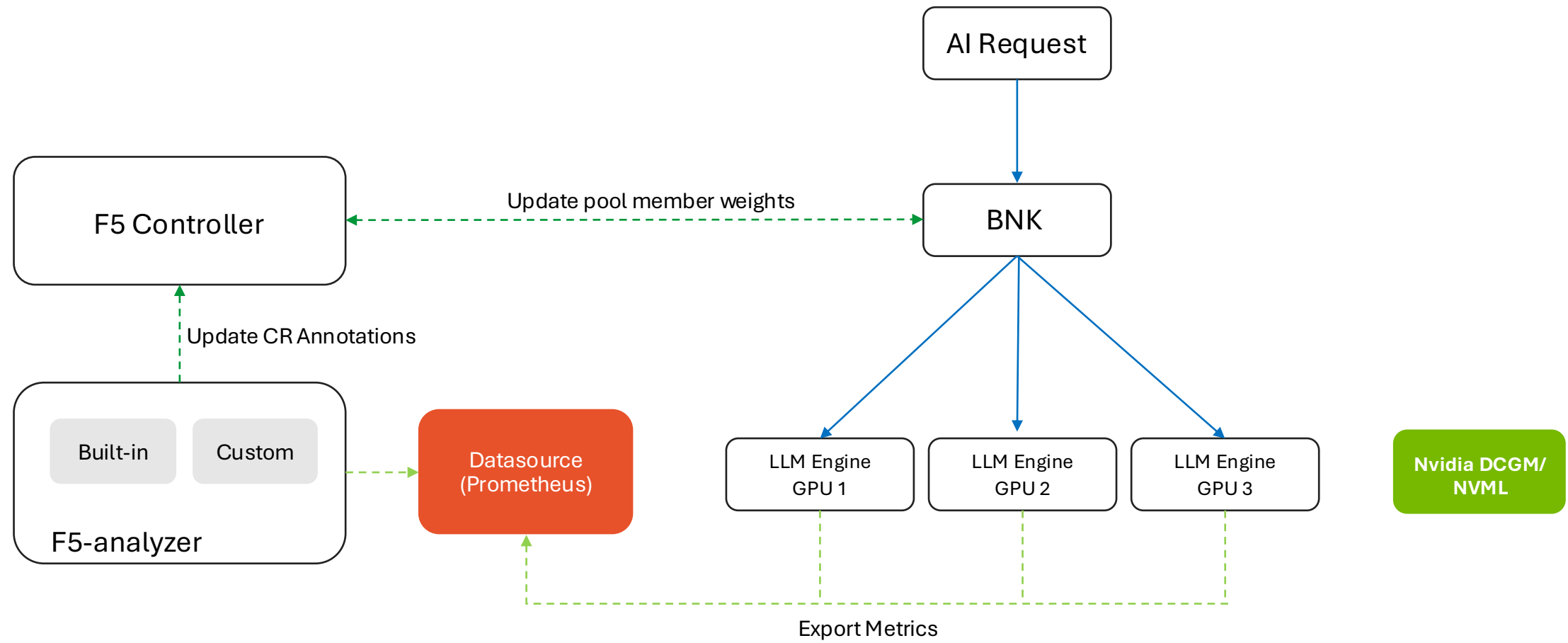
NVIDIA NIM Stats Integration

$$\left. \begin{array}{l} \text{Pending Requests} \\ \text{KV Cache} \\ \text{GPU Metrics} \\ \text{VRAM Utilization} \end{array} \right\} + = \text{Load Balance Rule}$$

2



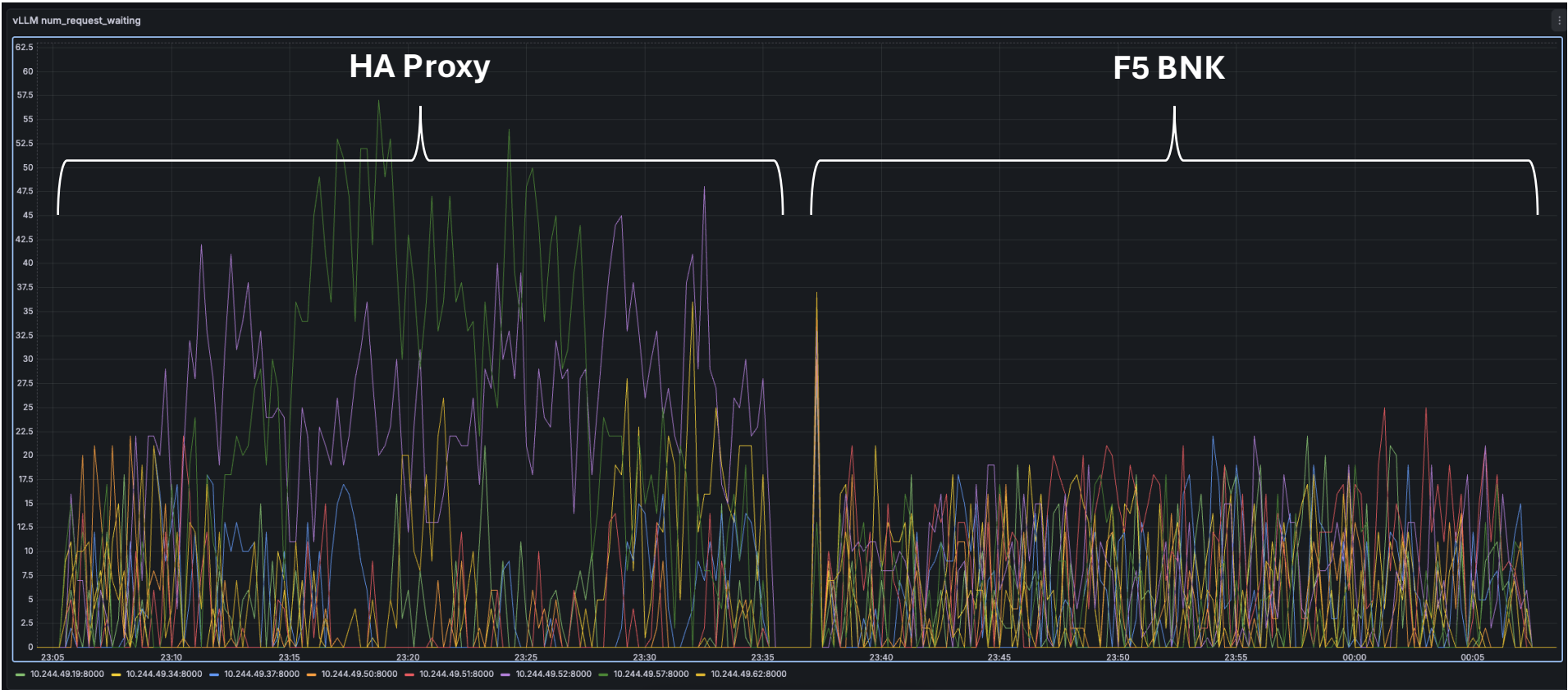
LLM Intelligent Load Balancing



Sample of queue numbers built up in vLLM during testings

Distribution of loads is uneven leads to erratic number in queue in different GPUs

Dynamic load distribution leads to smoother number in queue across GPUs



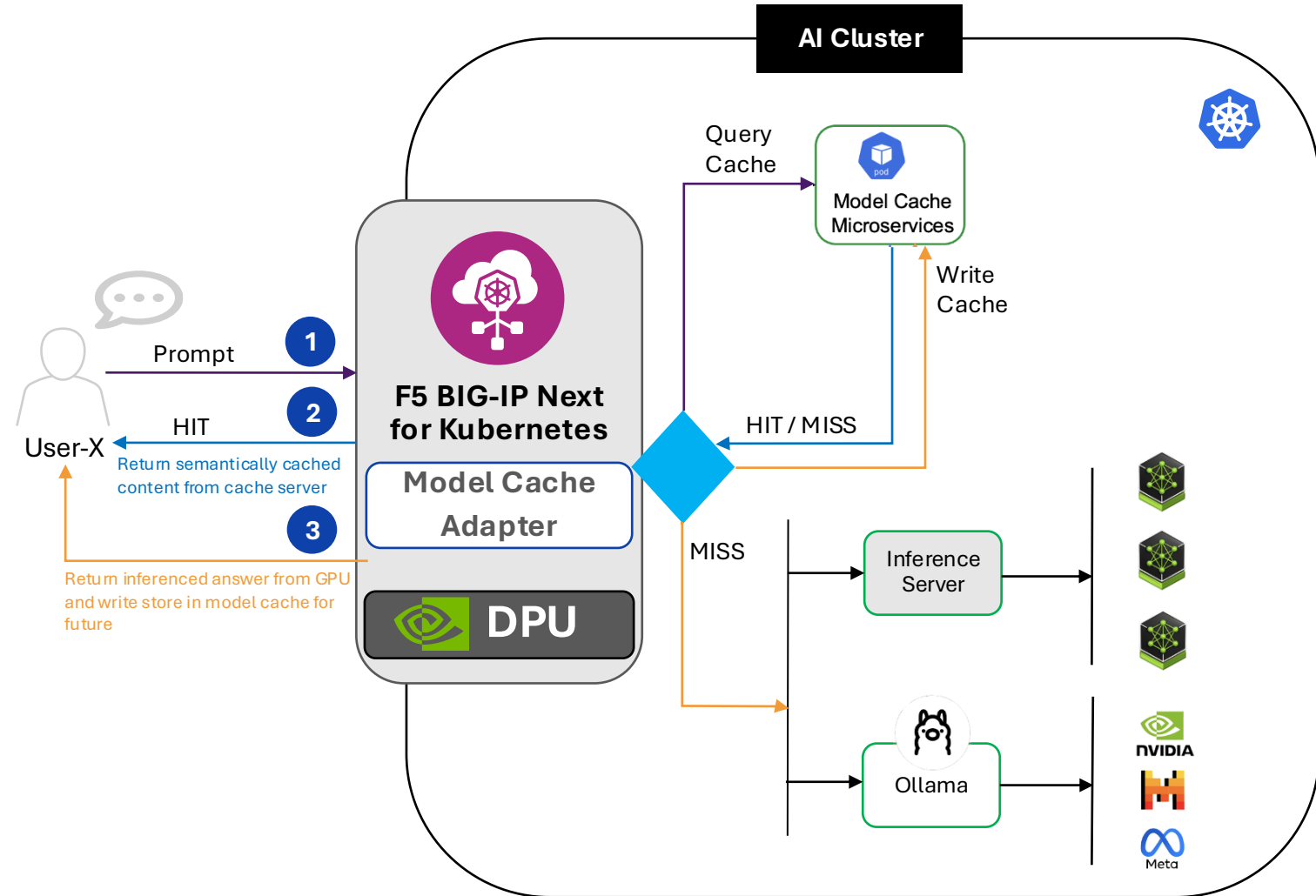
Semantic Model Caching accelerates inference responses by serving repeat or similar prompts directly from a model cache

Semantic Model Caching

- **Reduces GPU Load** by serving semantically similar responses from cache
- **Reduces latency** with instant cache hits for repeated or similar prompts
- **Improves token economics** and overall AI efficiency
- **Acts as a universal cache adapter**, integrating seamlessly with any inference backend

Flow:

- 1 **User Prompt → BNK:** BNK receives the prompt and checks the model cache
- 2 **Cache HIT → Instant Response:** If hit, BNK returns it instantly – no GPU compute.
- 3 **Cach MISS → Inference + Store:** BNK sends it to the inference server; GPU result is returned and written to cache.



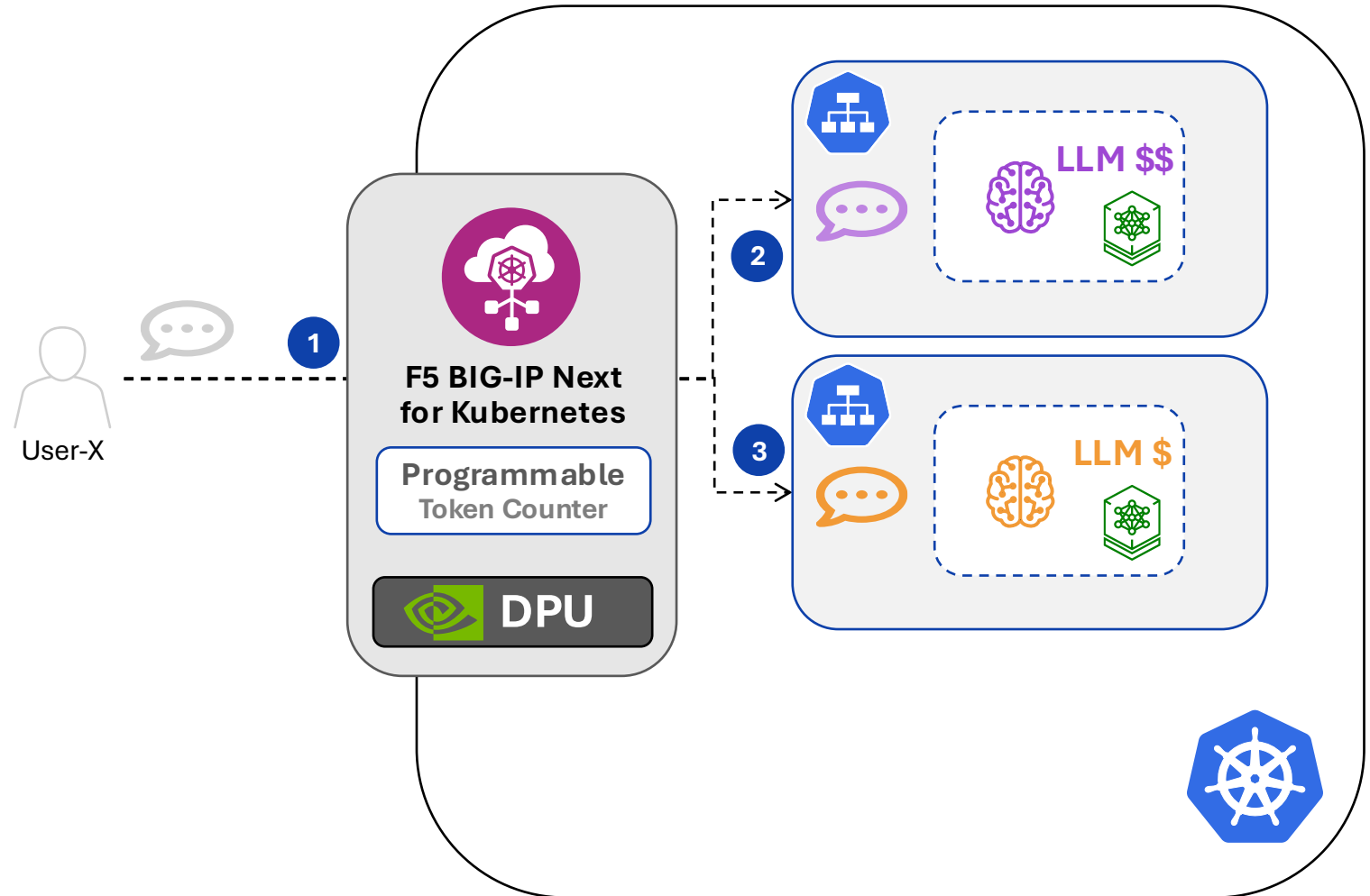
Token Enforcement – Programmable data plane enables the ability to track, manage, and control token usage

Programmable token enforcement for the delivery of...

- Token tracking
- Token Rate Limiting
- Token LLM routing
- Token hard limits

Example:

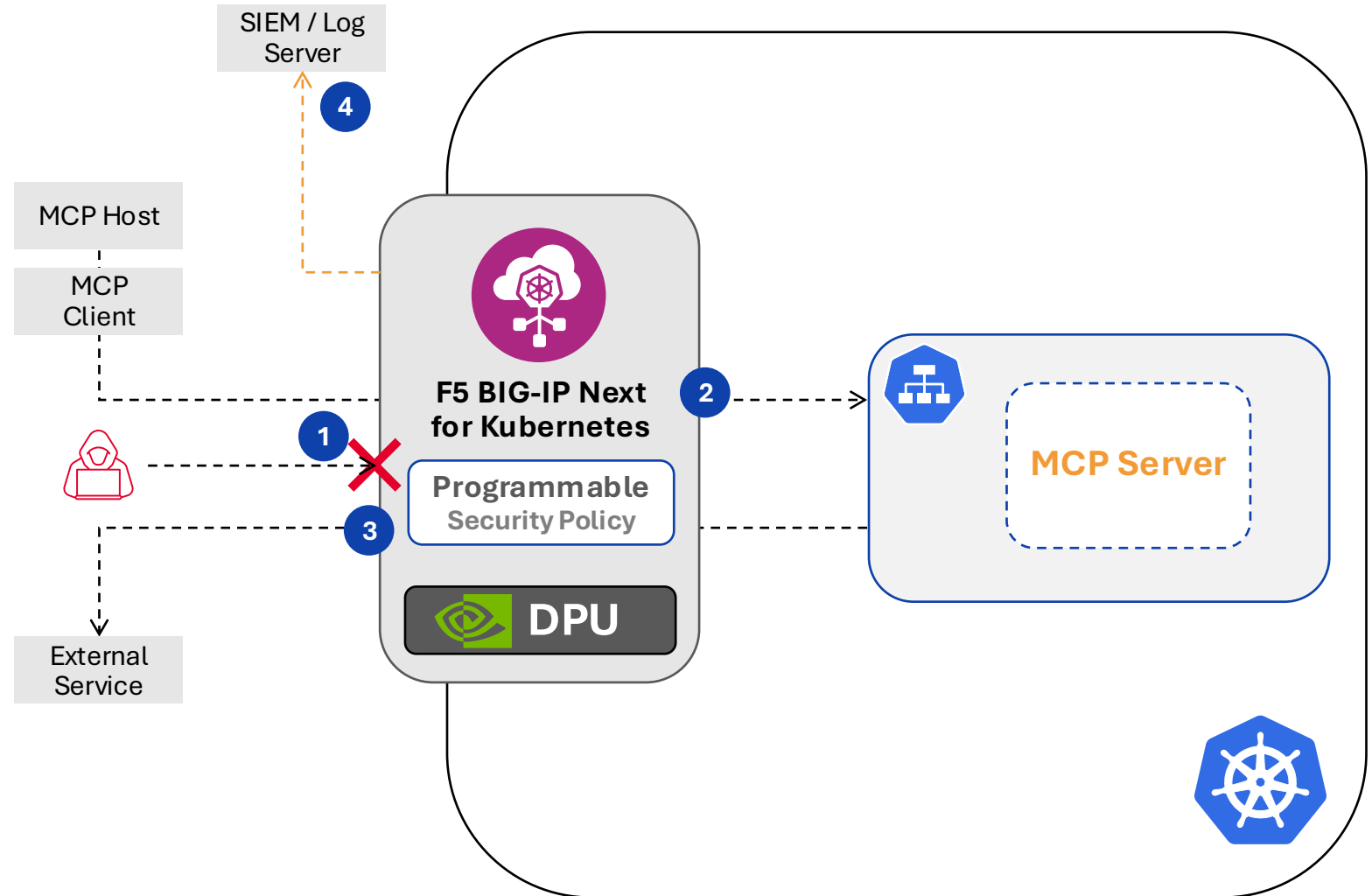
- 1 Token are counted for User-X BIG-IP
- 2 Requests are routed to more expensive LLM but policy is applied based on token limit
- 3 New request are routed to less expensive LLM



MCP Security L4 to L7 – Multi-layered programmable security for ingress and egress to MCP servers

MCP Servers are foundational for Agentic AI and security is required to secure data and protect against abuse

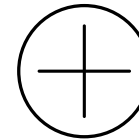
- 1 **1st layer of ingress security:** Edge Firewall & DDoS Protection
- 2 **2nd layer of ingress security:** Programmable enables Authorization (Tools) and sensitive data leakage prevention
- 3 **Egress security:** Facilitate trusted origin to external service
- 4 **Observability:** Audits and traceability



Intelligent and adaptable security solution protecting customer's critical AI Apps

NVIDIA DOCA Argus

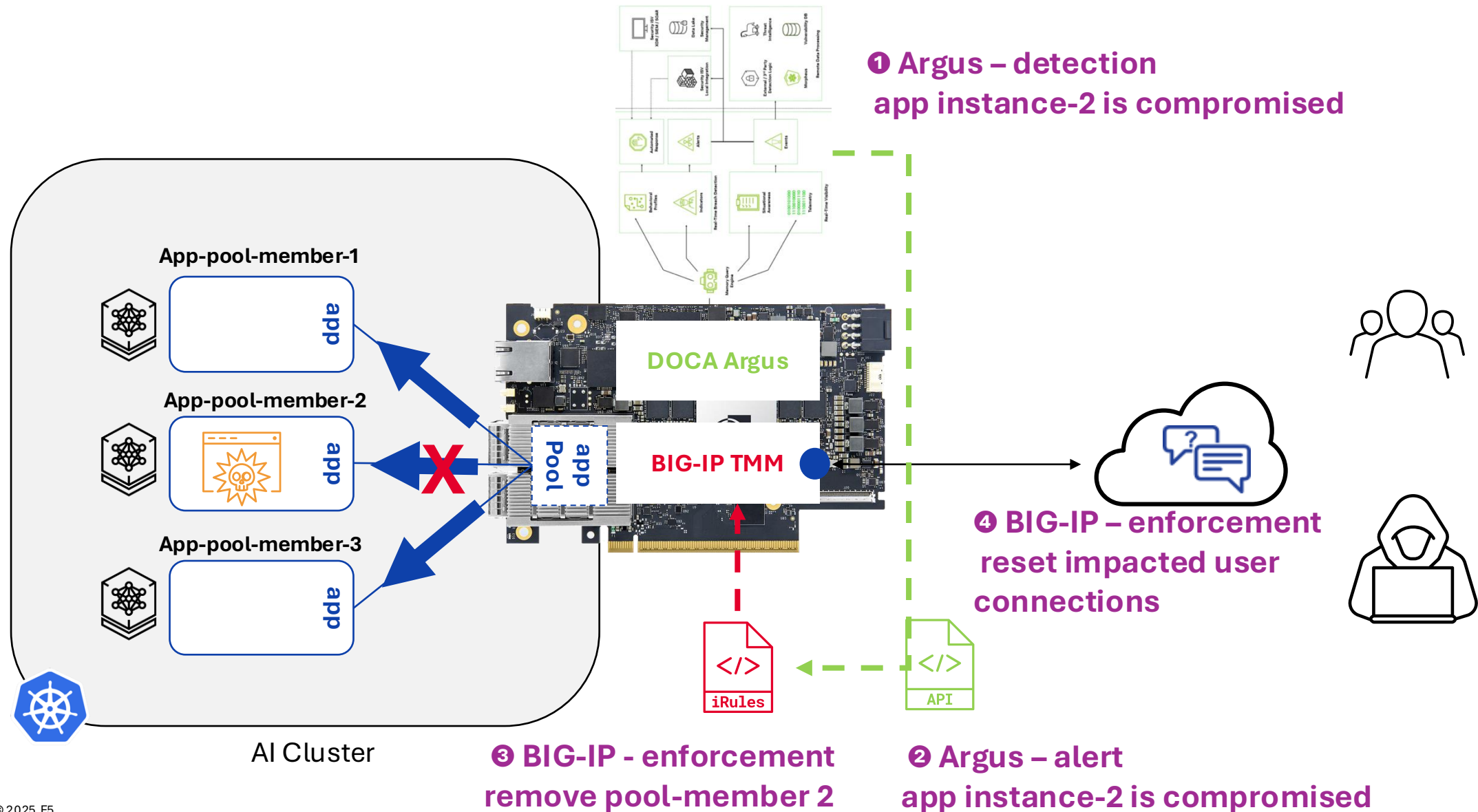
Cutting-edge cybersecurity framework designed to provide **real-time threat detection** for AI workloads



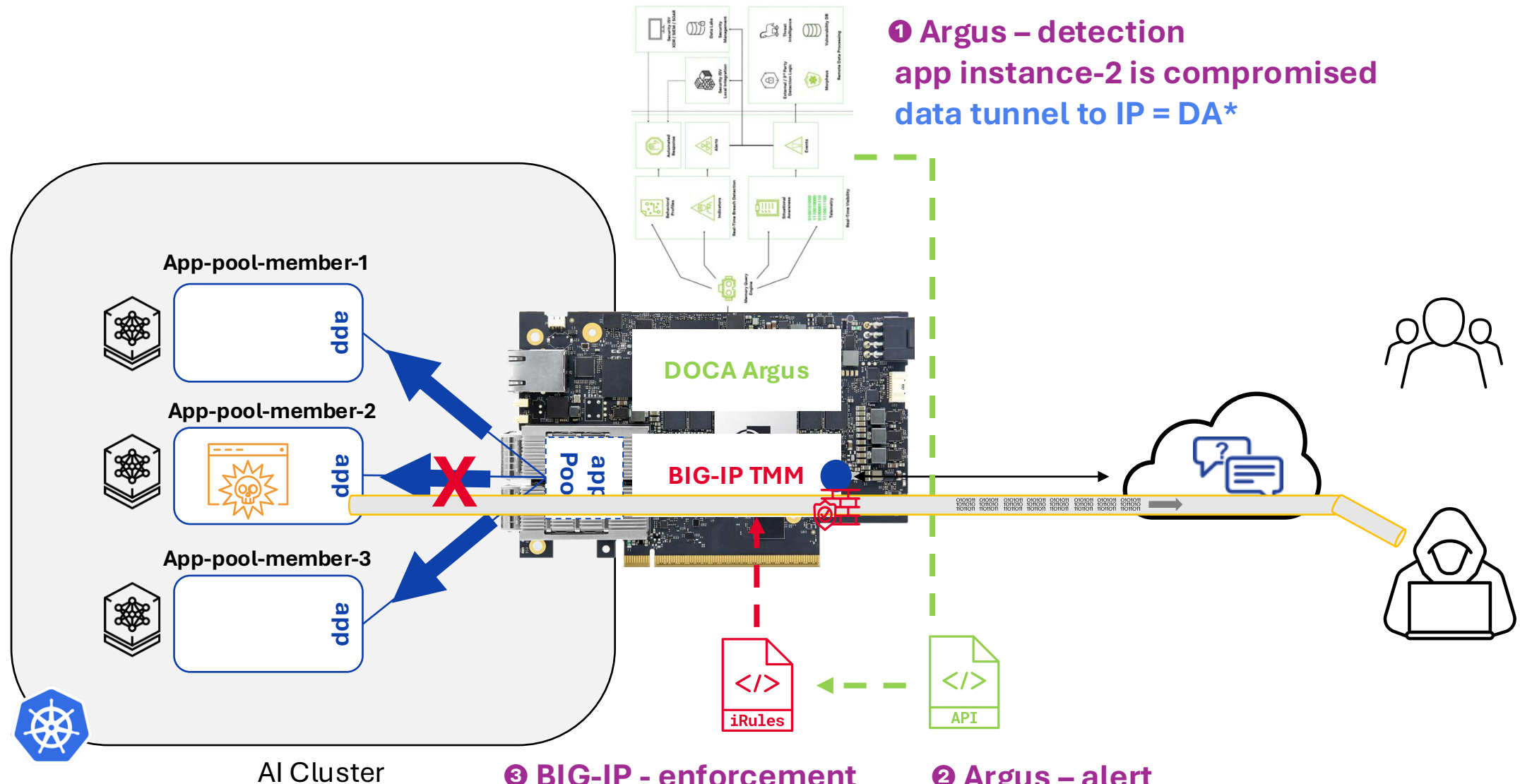
BIG-IP Next for K8s

High-performance app delivery and **adaptive programmable security** for AI workloads

Application Security Health-check, driving security policy



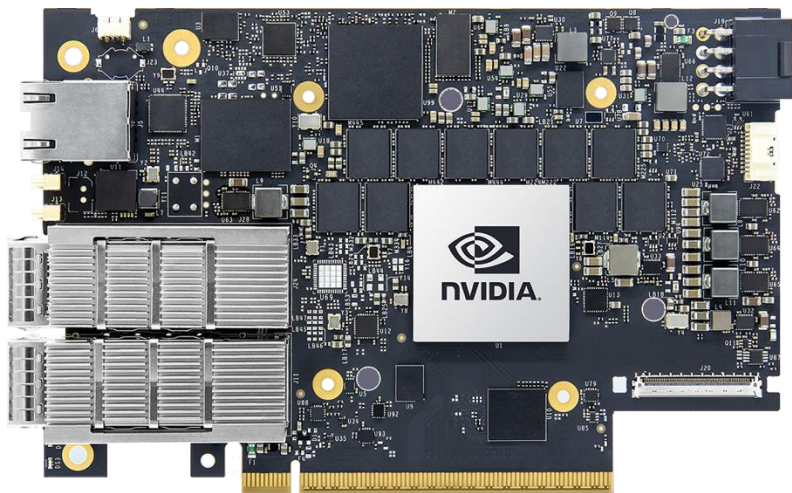
Data exfiltration detection & remediation



Offload Works: Customer-Proven, Featured by NVIDIA

DPU vs CPU for K8s Ingress – Real Cluster, Real Savings

Solution



18%

Higher HTTP Throughput:
running on a DPU vs a CPU-only
ingress.

11x

Better Time to First Byte:
means less user wait.

99%

Lower CPU Utilization:
more cores free for orchestration and
preprocessing.

190x

Better Energy Efficiency:
means greener clusters, lower OPEX.

“A real-world customer validated it – and NVIDIA
thought it mattered enough to feature it.”

CPU on dual Intel 6240R (24C, 165 W TDP)
DPU on BlueField-3 ARM (16C, ~44.8 W TDP)
CPU power assumes linear scaling with utilization

Source: Softbank, presented at an NVIDIA event.

Read NVIDIA's published write-up: <https://developer.nvidia.com/blog/turbocharging-ai-factories-with-dpu-accelerated-service-proxy-for-kubernetes/>

BIG-IP Next for Kubernetes on DPU delivers substantial gains across all critical AI performance metrics

**BNK on DPU delivers transformative AI performance gains—
Increasing token processing power by ~%50, accelerating TTFT by 85%, and cutting latency by 45%—
directly boosting GPU efficiency, token-economic value and overall service responsiveness**

+47%

↑

Higher Token Throughput

+86%

↑

Faster TTFT

+45%

↓

Lower Latency

	Output Tokens/sec	TTFT (ms)		End-to-End latency (ms)		Duration (min)
		p90		p90		
HAProxy on Host	1,349	60,528		85,610		60
Open-Source LB on Host	1,346	80,091		106,084		60
BIG-IP Next for Kubernetes on DPU	1,980	11,402		58,686		60
% Gain	+47%	BNK vs HAProxy	+81%	BNK vs HAProxy	+31%	
		BNK vs Open-Source LB	+86%	BNK vs Open-Source LB	+45%	

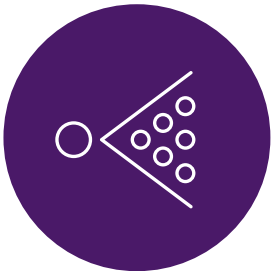
Test Case: 60 mins test run duration, 50 simultaneous users, with artificially overloading 4 out of 8 GPUs; 70b parameters



Independent validation: Turn AI infrastructure into output, not waste



Turn infrastructure into output: more tokens, faster responses, lower latency



40%

Higher token throughput
More tokens per second—
no model changes required



61%

Faster time to first token
Significantly improves
responsiveness for real-time
AI applications



34%

Lower latency
Stays fast and stable,
even under heavy load



BNK Forge

Fleet management

BNK Forge

INFRASTRUCTURE

OBSERVE

Command Center

BUILD

Deployment Stacks

Projects

Module Catalog

Helm Packages

OPERATE

Activity

Kubernetes

SETTINGS

Auth Templates

Workflows

System

Home

Infrastructure Management

Q Search...

SG-AI-Lab

sg-ai-lab-dpu0

All Namespaces

Q Search... (/)

Insights

Gateway Topology

Policy Gateway Map

AI Analyzers

Traffic Management

Security

Networking

Logging & Telemetry

System

Gateway Topology

Complete object graph showing how Gateways, Listeners, HTTPRoutes, Security Policies, iRules, and AI Analyzers connect together

3 Gateways

3 HTTPRoutes

3 Policies

2 iRules

2 VLANs

0 CNE Instances

Showing complete object graph for 3 gateways

Refresh

recipe-gateway f5-gateway-class

tenant1 VIP: 10.176.12.202

1 listener 1 route

http-listener HTTP :80

tenant1-http-gateway f5-gateway-class

tenant1 VIP: 10.176.12.201

1 listener 1 route

http-listener HTTP :80

tenant1-http-gateway-nginx f5-gateway-class

tenant1 VIP: 10.176.12.200

1 listener 1 route

http-listener HTTP :80

Data Plane TMM Networking

VLANs, interfaces, self-IPs, and CNE instance configuration

2 VLANs

sf-external External

sf-internal Internal

Static Routes 3

SNAT Pools 2

Egress 2

Logging & Telemetry 2

All systems operational

Version 2.7.14

Home

Infrastructure Management

Q Search...

SG-AI-Lab / sg-ai-lab-dpu0 / All Namespaces / Q Search... (/)

Insights

Gateway Topology

Policy Gateway Map

AI Analyzers

Traffic Management

Security

Networking

Logging & Telemetry

System

AI Load Balancing Analyzers

Intelligent traffic distribution for LLM inference workloads using F5BigAnalyzer CRDs

AI Load Balancing Analyzers

Viewing 1 analyzer for intelligent LLM traffic distribution

custom-traffic-analyzer f5-bnk

30s 1 application 1 data source Custom Script

Monitored Applications

Kind	Name	Namespace	API Group
HTTPRoute	l7-http-app	tenant1	gateway.networking.k8s.io

Data Sources

Name	Type	Endpoint	Auth
user-configured-prometheus	prometheus	http://prometheus-service.f5-analytics.svc.cluster.local:...	None

Script Configuration

Type: Custom

ConfigMap: custom-analyzer-scripts

Key: custom-plugin.py

Status Details

Created: 23d

Conditions:

Accepted (Accepted)

Programmed (Programmed)

